Code-Layout, Readability and Reusability

Date	03 October 2026
Team ID	TEAM-ID-TO-BE-FILLED
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4 spaces in Python, 2 in HTML/JS/CSS
2	Proper File Structure	Files and folders are logically organized	Yes	`main.py` routes, `gemini_utils.py` AI service, `auth.py`, `models.py`, `database.py`, `config.py`, `templates/`, `static/`, `tests/` (pytest only), `tools/` (Gemini check, load test, demo seeding, docs build)
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. `remaining_budget`, `shopping_links`, `blacklisted_tokens`
4	Function / Method Names	Functions are descriptively named	Yes	e.g. `get_home_recommendations`, `build_shopping_links`, `save_upload_file`
5	Code Comments	Inline and block comments explain logic	Yes	Module docstrings and comments for non-obvious logic
6	Modular Design	Code is split into reusable functions/modules	Yes	One module per concern
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared finalize/trim/link helpers serve all three planners; one JS renderer for results and history
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Validation errors (422), auth (401), uploads (413/415), AI failures fall back to default plans

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	`build_shopping_links()` and `PLATFORM_URLS`	Python	Home, Party and Jewelry planners, venue suggestions	High
2	`_finalize_breakdown()` / `_trim_to_budget()`	Python	Home, Party and Jewelry validation	High
3	`_generate_json()` Gemini client with model chain	Python	All three planners (text and image)	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
4	`renderPlan()` JS renderer	JavaScript	Planner result pages and history detail modal	High
5	`base.html` layout	Jinja2	All 8 pages	High
6	`auth.get_current_active_user` dependency	Python / FastAPI	Every protected route	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of routes, services, models and storage
Readability	4	Descriptive names and short functions; `gemini_utils.py` is the longest file
Reusability	5	Shared helpers across planners and pages
Documentation / Comments	4	Docstrings, README and phase documents
Overall Score	**4.5**	